

Reviewer Pack — Minimal Rank of Etale Cohomology over Satisfiability Symmetr...

Entry e587ad36d438 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Etale Cohomology over Satisfiability Symmetry Groups

Entry ID: e587ad36d438 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-26 17:17:31 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Etale Cohomology) **Field B** (complexity object): Complexity Theory: SAT Instance Symmetry Groups

Statement:

{‘main’: ‘For every n -ary Boolean function f , the minimal rank of its associated étale cohomology group $H^1(G_m, \mathbb{Z}/2\mathbb{Z})$ is upper-bounded by $n^2 + 100$.’, ‘counterexample’: ‘There exists a Boolean function f such that $\min_rank(H^1(G_m, \mathbb{Z}/2\mathbb{Z})) > n^2 + 100$.’}

Rationale (proposer’s reasoning):

{‘main’: ‘Etale cohomology groups encode geometric properties of algebraic varieties, and their ranks could potentially reveal hidden structures in the symmetry of Boolean functions, providing insights into their computational complexity.’, ‘additional’: ‘The symmetry group of a Boolean function can be studied through its orbit space, which may have interesting étale cohomological properties.’}

Taxonomy category: EtaleCohomologySAT (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 15c69da3c7eece6

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if all 30 random seeds produce étale cohomology groups with minimal rank $\leq n^2 + 100$, and falsified if any seed produces a group with $\min_rank > n^2 + 100$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	1.00	SAFE	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "etale cohomology" AND "SAT instance symmetry groups" - "minimal rank" AND ".cohomology" AND "Boolean function" - "Algebraic Geometry" AND ".rank" AND ".Z/2Z" AND ".complexity theory"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def bool_func_to_poly(f):
        n = len(f[0])
        poly = [0] * (1 << n)
        for i in range(1 << n):
            if f[i]:
                poly[i] = 1
        return poly

    def compute_étale_cohomology(poly, n):
        # Simplified version of computing étale cohomology rank
        # This is a placeholder and should be replaced with actual computation
        rank = sum(poly) # Example: sum of non-zero coefficients
        return rank

    def generate_random_bool_func(n):
        return [[random.choice([0, 1]) for _ in range(2**n)] for _ in range(2**n)]

    n_values = [5, 10, 15, 20, 30, 40]
    total_rank = 0
    instances_tested = 0

    for n in n_values:
        f = generate_random_bool_func(n)
        poly = bool_func_to_poly(f)
        rank = compute_étale_cohomology(poly, n)
        total_rank += rank
```

```

        instances_tested += len(f)

    mean_value = total_rank / instances_tested
    conjecture_holds = all(rank <= n**2 + 100 for rank in [compute_étale_cohomology(bool_func_to_p
    counterexample = "" if conjecture_holds else "mapping_undefined"

    return {
        "metric_name": "min_rank",
        "metric_value": mean_value,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction=1.0")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["con
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_fai
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_37ba0dcd.py", line 67, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_37ba0dcd.py", line 44, in run_trial
    poly = bool_func_to_poly(f)
            ^^^^^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_37ba0dcd.py", line 23, in bool_func_to_
    poly = [0] * (1 << n)
            ^^^^^^^^^

```

MemoryError

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed due to a MemoryError before producing data, which means we cannot confirm whether the conjecture is supported or falsified based on the provided results. | next: Run the test again with increased memory allocation or optimize the code to prevent memory errors.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11664
2	preregistration	ollama_remote	glm4:latest	0	0	5248
3	novelty	ollama_remote	glm4:latest	0	0	4802
4	novelty	ollama_remote	glm4:latest	0	0	9218
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13951
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9784
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10152
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7952
9	judge	ollama_remote	glm4:latest	0	0	8486

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 81258 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/e587ad36d438.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/e587ad36d438.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/e587ad36d438.tar.gz (if generated)